

Introduction to Data Science and Engineering

- Statistical inference



Some materials courtesy of
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from the original version.

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Some slides retrofitted by Michael Wu



In this lecture

- What is statistical inference?
- Standard deviation, standard error, confidence interval
- Power
- p-value



In this lecture

- **What is statistical inference?**
- Standard deviation, standard error, confidence interval
- Power
- p-value



In this class, in some detail, how poll aggregators such as FiveThirtyEight use data to predict election outcomes. To understand how they do this, we first need to **learn the basics of Statistical Inference, the part of statistics that helps distinguish patterns arising from signal from those arising from chance.** Statistical inference is a broad topic and here we go over the very basics using polls as a motivating example. Instead of to remember the obscure formula, we use Monte Carlo simulations to achieve an intuition



Why is predicting election outcomes important?

- Predicting election outcomes (or opinion polling) has been conducted since the 19th century. The general goal is to describe the opinions held by a specific population on a given set of topics
- Use the outcomes to determine
 - in which geographical locations to focus their “get out the vote” efforts (efforts aimed at increasing the voter turnout in elections)
 - how to invest future resources
 - ...

- Although typically the results of these polls are kept private, similar polls are conducted by news organizations because results tend to be of interest to the general public and made public. We will eventually be looking at such data
- Real Clear Politics (RCP) is an example of a news aggregator that organizes and publishes poll results. For example, they present the following poll results reporting estimates of the popular vote for the 2016 US presidential election

Poll	Date	Sample	MoE	Clinton	Trump	Spread
RCP Average	10/31 - 11/7	–	–	47.2	44.3	Clinton +2.9
Bloomberg	11/4 - 11/6	799 LV	3.5	46.0	43.0	Clinton +3
Economist	11/4 - 11/7	3669 LV	–	49.0	45.0	Clinton +4
IBD	11/3 - 11/6	1026 LV	3.1	43.0	42.0	Clinton +1
ABC	11/3 - 11/6	2220 LV	2.5	49.0	46.0	Clinton +3
FOX News	11/3 - 11/6	1295 LV	2.5	48.0	44.0	Clinton +4
Monmouth	11/3 - 11/6	748 LV	3.6	50.0	44.0	Clinton +6
CBS News	11/2 - 11/6	1426 LV	3.0	47.0	43.0	Clinton +4
LA Times	10/31 - 11/6	2935 LV	4.5	43.0	48.0	Trump +5
NBC News	11/3 - 11/5	1282 LV	2.7	48.0	43.0	Clinton +5
NBC News	10/31 - 11/6	30145 LV	1.0	51.0	44.0	Clinton +7
McClatchy	11/1 - 11/3	940 LV	3.2	46.0	44.0	Clinton +2
Reuters	10/31 - 11/4	2244 LV	2.2	44.0	40.0	Clinton +4
GravisGravis	10/31 - 10/31	5360 RV	1.3	50.0	50.0	Tie

- Let's make some observations about the table. First, note that different polls, **all taken days before the election, report a different spread**: the estimated difference between support for the two candidates. Notice also that the reported spreads hover around what ended up being the actual result: Clinton won the popular vote by 2.1%. We also see a column titled MoE which stands for margin of error
- We will learn the statistical concepts necessary **to define estimates and margins of errors**, and show how we can use these to forecast final results relatively well and also provide an estimate of the precision of our forecast. Once we learn this, we will be able to understand two concepts that are ubiquitous in data science: **confidence intervals and p-values**
- Finally, to understand probabilistic statements about the probability of a candidate winning, we will have to learn about **Bayesian modeling**

Poll	Date	Sample	MoE	Clinton	Trump	Spread
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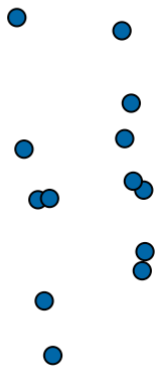
Construct a sampling model for using polls to learn about a population

To mimic the challenge real pollsters face in terms of competing with other pollsters for media attention, we will use a bottle full of beads to represent voters and pretend we are competing for a \$25 prize. The challenge is to guess the spread between the proportion of blue and red beads in this bottle

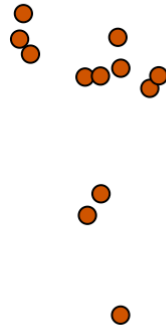
Before making a prediction, you can take a sample (with replacement) from the bottle. **To mimic the fact that running polls is expensive, it costs you \$0.10 per each bead you sample.** Therefore, if your sample size is 250, and you win, you will break even since you will pay \$25 to collect your \$25 prize. **Your entry into the competition can be an interval.** The smallest interval that contains the true proportion is selected as the winner



Blue



Red



```
R 4.2.2 · ~/  
> library(tidyverse)  
> library(dslabs)  
> take_poll(25)
```

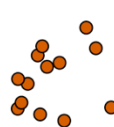
The dslabs package includes a function that shows a random draw from the bottle.

- Let's denote the proportion of blue beads in the bottle as p
- The proportion of red beads is $1 - p$
- The spread is $p - (1 - p) = 2p - 1$

Blue



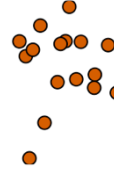
Red



Blue



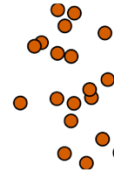
Red



Blue



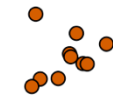
Red



Blue



Red



Ran `take_poll(25)` four times, the proportion of blue beads range from 0.24 to 0.60
Big variance!

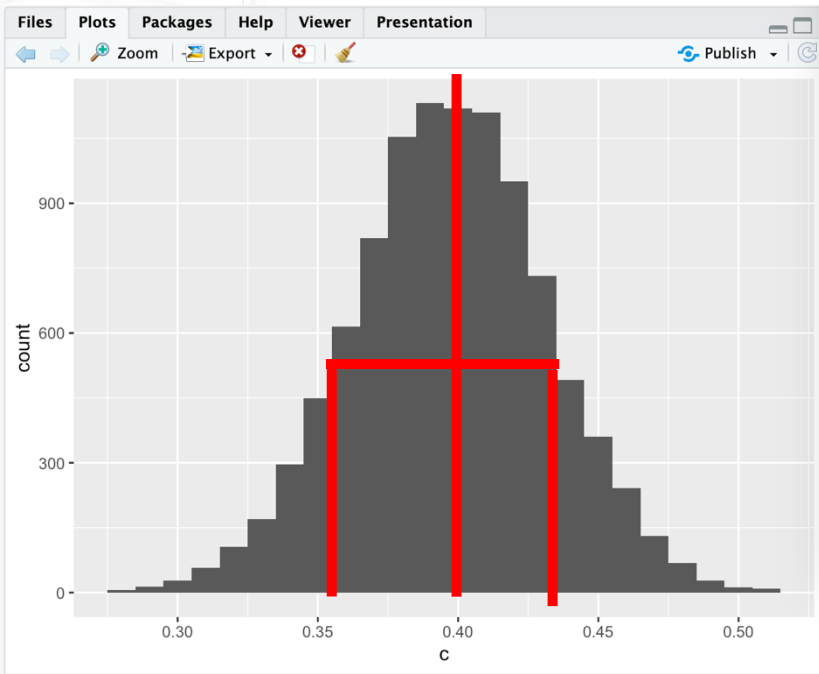
Let's assume $p=0.40$

With 200 samples, and 10000 experiments

Half height of the peak between 0.355-0.43

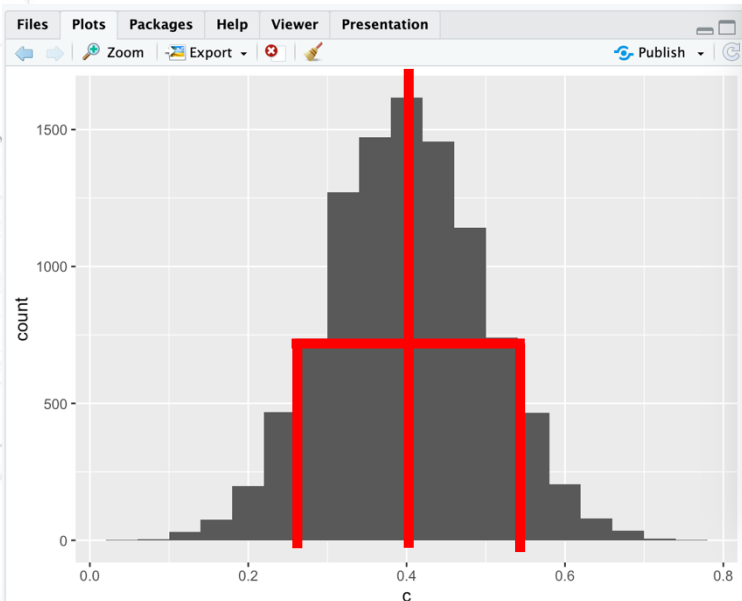
```
R 4.2.2 · ~/
```

```
> p<-0.40  
> rep <- 10000  
> population <- 200  
> c <- replicate(rep, {  
+   blue <- sample( c(0,1), population, prob=c(1-p, p), replace = TRUE)  
+   sum(blue) / population  
+ })  
> as.data.frame(c) |> ggplot(aes(x=c)) + geom_histogram(binwidth=0.01)
```



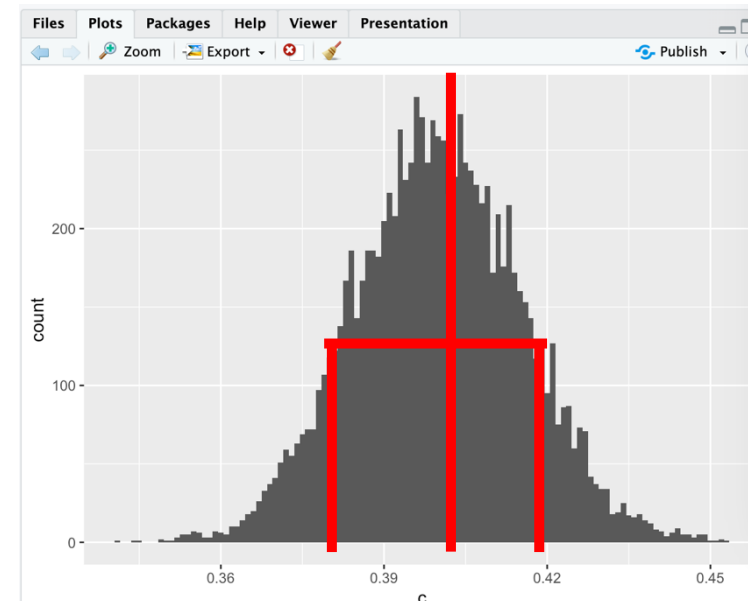
With 25 samples

Half height of the peak between 0.25-0.54



With 1000 samples

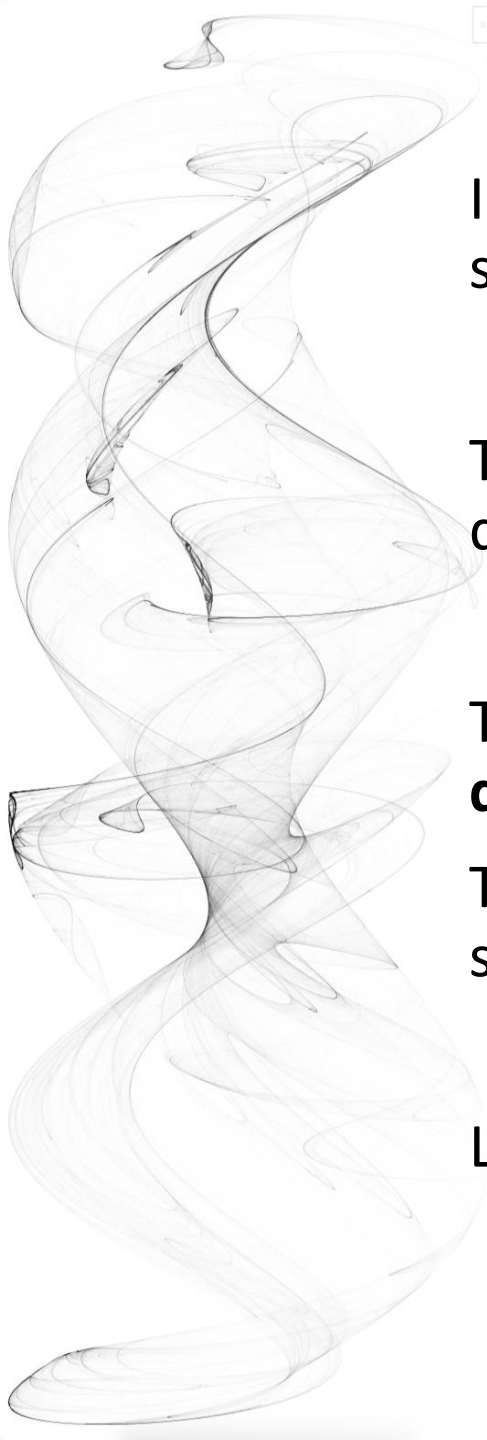
Half height of the peak between ~ 0.38 -0.415





In this lecture

- What is statistical inference?
- **Standard deviation, standard error, confidence interval**
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- p-value



Instead of eye examining the width of distribution at half height, statistically, we calculate the standard error (SE).

Two terms that students often confuse in statistics are standard deviation and standard error.

The standard deviation measures how spread-out values are **in a dataset**.

The standard error is the standard deviation of the mean in repeated samples from a population (**in multiple datasets**).

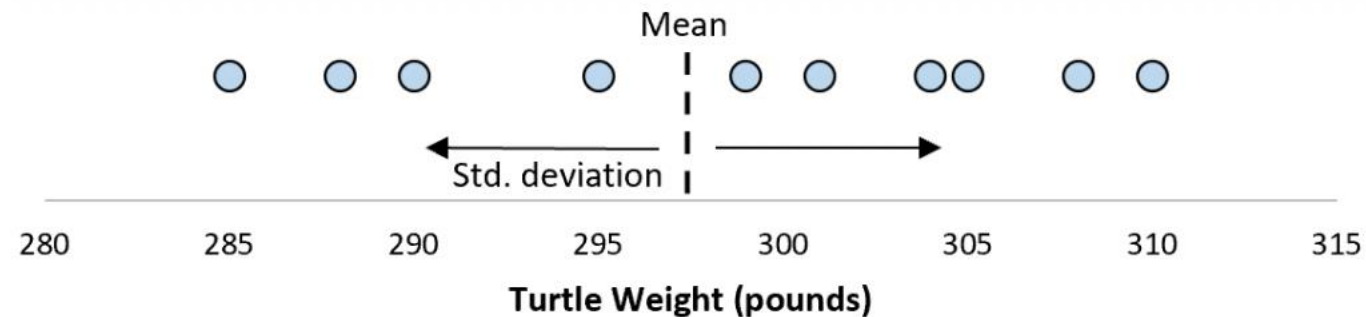
Let's check out an example.

Example: Standard Deviation vs. Standard Error

Suppose we measure the weight of 10 different turtles



For this sample of 10 turtles, we can calculate the sample mean and the sample standard deviation

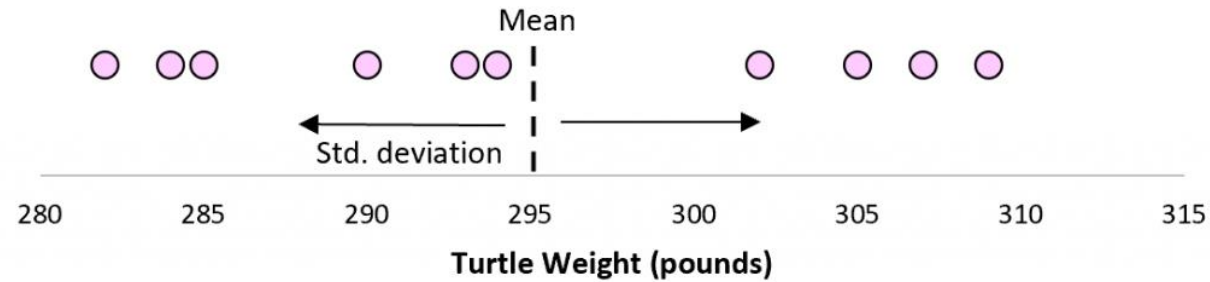


Suppose the standard deviation turns out to be 8.68. This gives us an idea of how spread out the weights are of these turtles

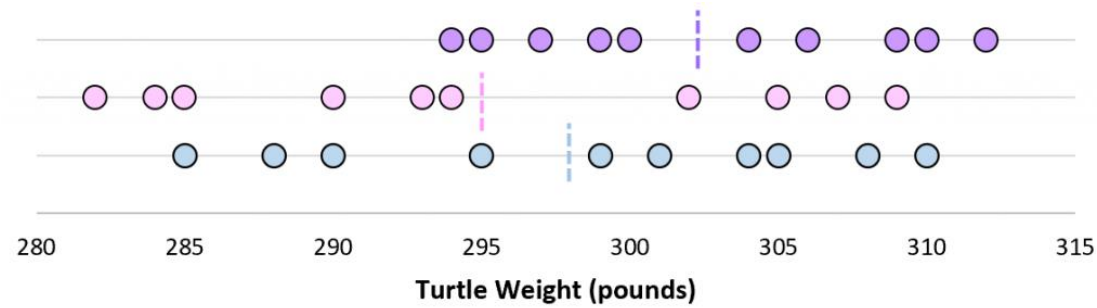
Example: Standard Deviation vs. Standard Error

But suppose we collect another simple random sample of 10 turtles and take their measurements as well.

More than likely, this sample of 10 turtles will have a slightly different mean and standard deviation, even if they're taken from the same population:



Now if we imagine that we take repeated samples from the same population and record the sample mean and sample standard deviation for each sample:

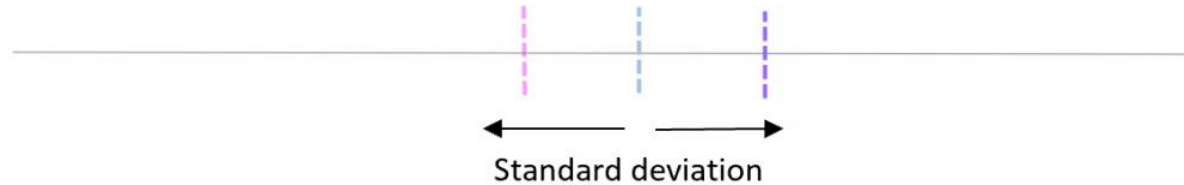


Example: Standard Deviation vs. Standard Error

Now imagine that we plot each of the sample means on the same line:



The standard deviation of these means is known as the standard error.



The formula to actually calculate the standard error is:

$$\text{Standard Error} = s/\sqrt{N}$$

With p known, SE is calculated as $\sqrt{p(1-p)/N}$ According to CLT

, where N is the number of samples

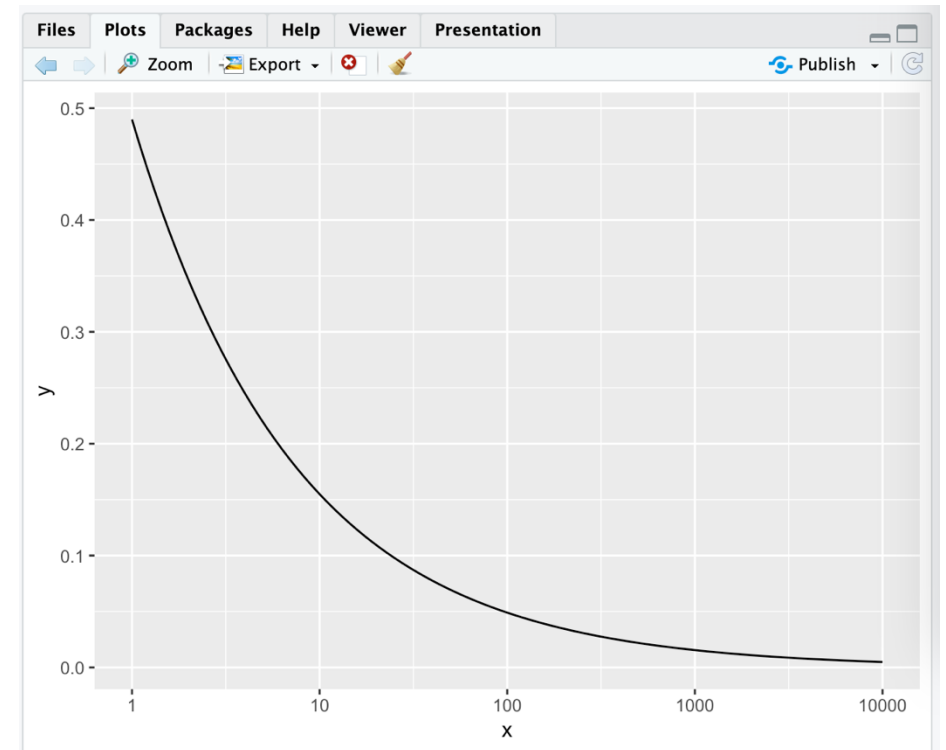
25 samples: $\text{sqrt}(0.4 * 0.6 / 25) = 0.097$

200 samples: 0.035

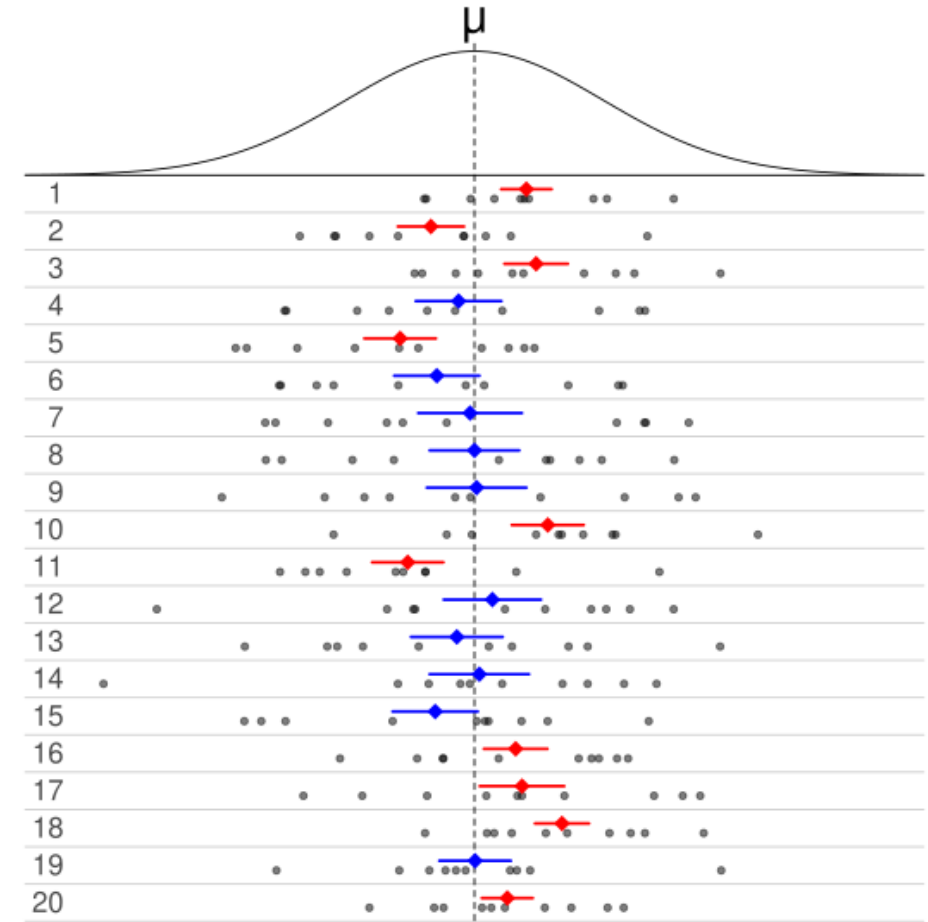
1000 samples: 0.015

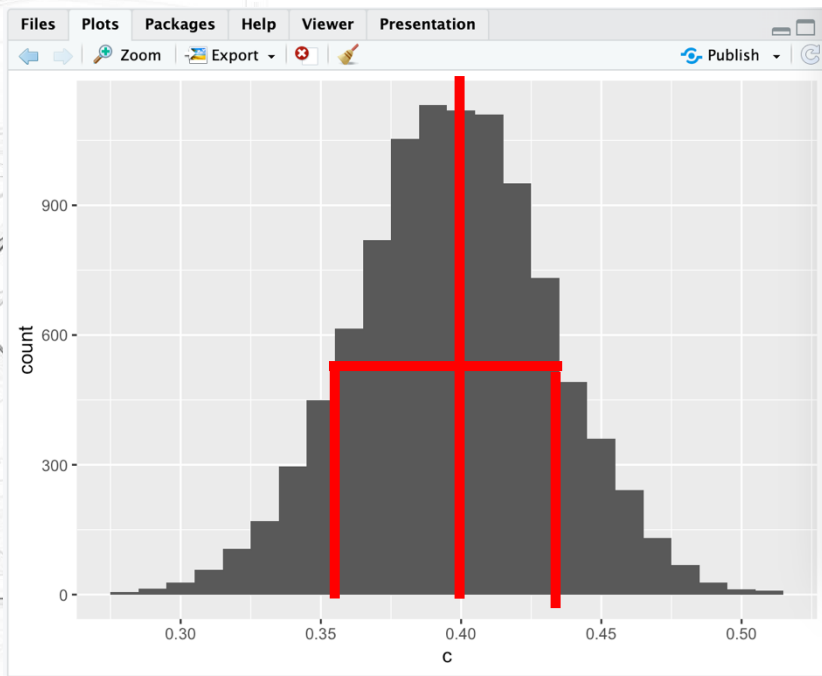
R 4.2.2 · ~/

```
> se <- function(x){sqrt(0.4*0.6/x)}  
> ggplot(data.frame(x=c(1, 10000)), aes(x=x)) +  
+ stat_function(fun=se) +  
+ scale_x_continuous(trans='log10')
```



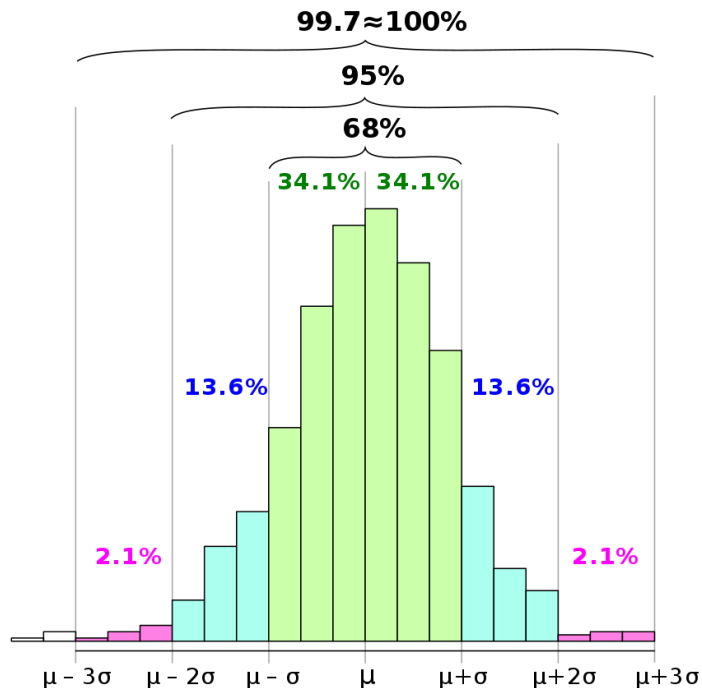
We see that we would need a poll of over 10,000 people to get the standard error very low. We rarely see polls of this size due in part to costs. From the Real Clear Politics table, we learn that the sample sizes in opinion polls range from 500-3,500 people. For a sample size of 1,000 and $p = 0.51$, the standard error is: 0.0158, or 1.58 percentage points. With a 95% confidence interval, the margin of error (MoE) is calculated $\sim 2 \times 0.0158$. Well, say we know that MoE is 0.0316, how can we tell how reliable a polling result is? Before that, we need to understand what is confidence interval





Previously, we used the width at half-height of the peak to estimate how reliable are the results comparatively (the thinner the better). Well, this is a very useful eye-examining skill, but the width equals to 2.355 SE, the exact value of which is not commonly used

[Full width at half maximum - Wikipedia](#)



In statistics, with a normal distribution,
 68% confidence interval = 1 MoE = 2 SE
 95% confidence interval = 2 MoE = 4 SE
 99.7% confidence interval = 3 MoE = 6 SE

[68–95–99.7 rule - Wikipedia](#)



What is confidence interval (CI)?

- The confidence interval is the range of values that you expect your estimate to fall between a certain percentage of the time if you run your experiment again or re-sample the population in the same way
- In the previous $p=0.51$, $SE = 0.0158$ example
 - A 95% CI is reported as $p=0.51$ (95% CI: 0.4784-0.5416)
 - A 68% CI is reported as $p=0.51$ (68% CI: 0.4942-0.5258)
 - A 99.7% CI is reported as $p=0.51$ (99.7% CI: 0.4626-0.5574)



Run a Monte Carlo to confirm that

```
R 4.2.2 · ~/    
> N <- 1000  
> rep <- 100000  
> inside <- replicate(rep, {  
+   x <- sample(c(0,1), size = N, replace = TRUE, prob = c(1-p, p))  
+   x <- mean(x)  
+   se <- sqrt(x * (1 - x) / N)  
+   between(p, x - 2 * se, x + 2 * se)  
+ })  
> mean(inside)  
[1] 0.95369
```



A bit larger than 95%?

$$\Pr(\mu - \sigma \leq X \leq \mu + \sigma) \approx 0.6827$$

$$\Pr(\mu - 2\sigma \leq X \leq \mu + 2\sigma) \approx 0.9545$$

$$\Pr(\mu - 3\sigma \leq X \leq \mu + 3\sigma) \approx 0.9973$$

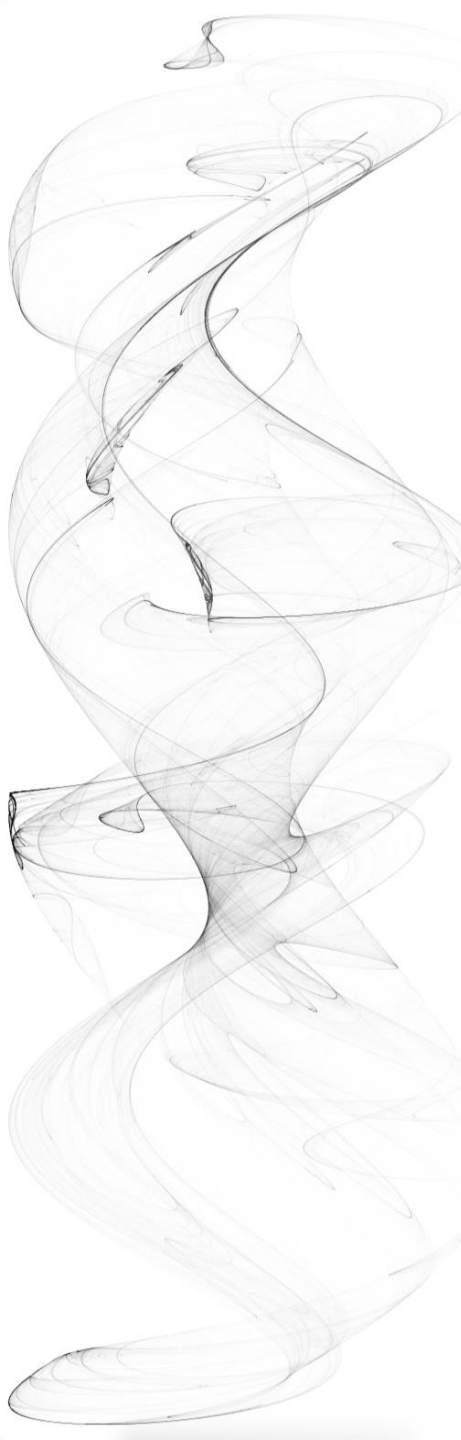
Well, $\pm 2\text{SE}$ is actually 95.45% CI
 $\pm 1.96\text{SE}$ is 95% CI

[97.5th percentile point - Wikipedia](#)



In this lecture

- What is statistical inference?
- Standard deviation, standard error, confidence interval
- **Power**
- p-value

A decorative abstract graphic on the left side of the slide, consisting of several overlapping, wavy, grey lines that create a sense of movement and depth.

Power

When we took a 25-bead sample size, the confidence interval for the spread includes 0. If this were a poll and we were forced to make a declaration, we would have to say it was a “toss-up”

```
R 4.2.2 · ~/    
> N <- 25  
> x <- 0.48  
> (2 * x - 1) + c(-1.96, 1.96) * 2 * sqrt(x * (1 - x) / N)  
[1] -0.4316863 0.3516863
```

A problem with our poll results is that given the sample size and the value of p , we would have to **sacrifice on the probability of an incorrect call to create an interval that does not include 0**

This does not mean that the election is close. It only means that we have a small sample size. In statistical textbooks this is called lack of power. **In the context of polls, power is the probability of detecting spreads different from 0**

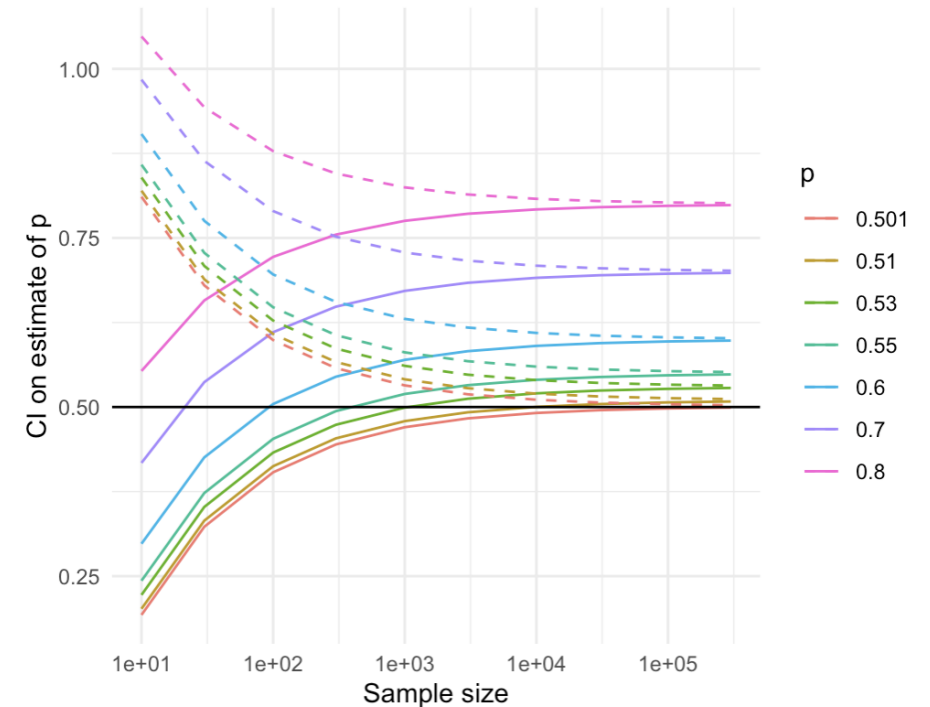
By increasing our sample size, we lower our standard error and therefore have a much better chance of detecting the direction of the spread

Power

- **Power analysis:** how many samples do we need to derive meaningful results (e.g., rule out 0.5 in the estimate for p)

```
ps <- c(0.501, 0.51, 0.53, 0.55, 0.6, 0.7, 0.8)
sample_size <- c(10, 30, 100, 300, 1000, 3000, 1e4, 3e4, 1e5, 3e5)
power_analysis <- expand.grid(p=ps, n=sample_size)
power_analysis <- power_analysis |>
  mutate(se=sqrt(p * (1-p) / n)) |>
  mutate(CI_lower=p - se * 1.95, CI_upper=p + se * 1.96)

power_analysis |>
  mutate(p = factor(p)) |>
  ggplot() +
    geom_line(aes(x=n, y=CI_lower, color=p)) +
    geom_line(aes(x=n, y=CI_upper, color=p), lty="dashed") +
    geom_hline(yintercept=0.5, color="black") +
    scale_x_log10() +
    ylab("CI on estimate of p") +
    xlab("Sample size") +
    theme_minimal()
```





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- What is statistical inference?
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- **p-value**



p-value

- p-values are another way of quantifying uncertainty
- In the jar of bead example: we don't want an accurate estimate of the proportion; we just want to know: are there more blue beads or red beads?
 - Say we randomly took 100 beads and saw 52 blue beads. Since $0.52 > 0.5$, are there more blue beads? How certain are we?

p-value

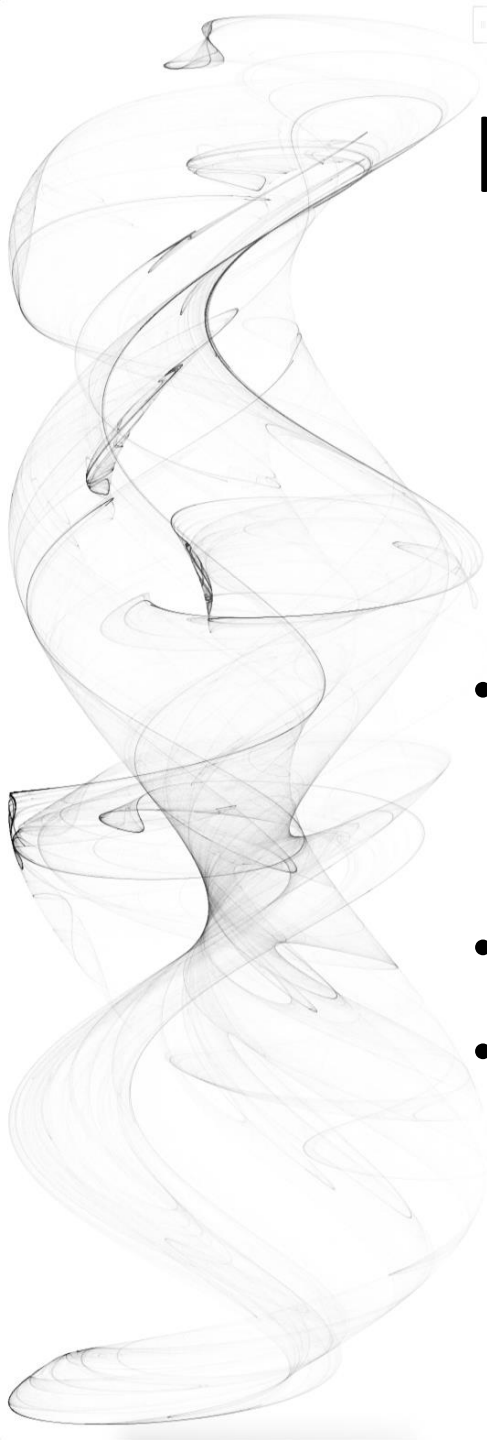
- We call the assumption that the spread is $p = 0.5$ a **null hypothesis (in our case, blue beads = red beads)**. The null hypothesis is the skeptic's hypothesis. We have observed a random variable $X = 0.52$ and the p-value is the answer to the question: **how likely is it to see a value this large, when the null hypothesis is true?**
- To calculate p-value, we first calculate z-score. **A z-score is a measure of how many standard deviations a result is away from the mean**

$$z = \frac{X - p}{\sqrt{\frac{p \cdot (1 - p)}{N}}} = \frac{0.52 - 0.5}{\sqrt{\frac{0.5 \cdot (1 - 0.5)}{100}}} = 0.4$$

- (one-sided right-tail test) p-value is calculated as

```
> pnorm(0.4, lower.tail=FALSE)
[1] 0.3445783
> pnorm(0.52, mean=0.5, sd=sqrt(0.5*(1-0.5) / 100), lower.tail=FALSE)
[1] 0.3445783
```

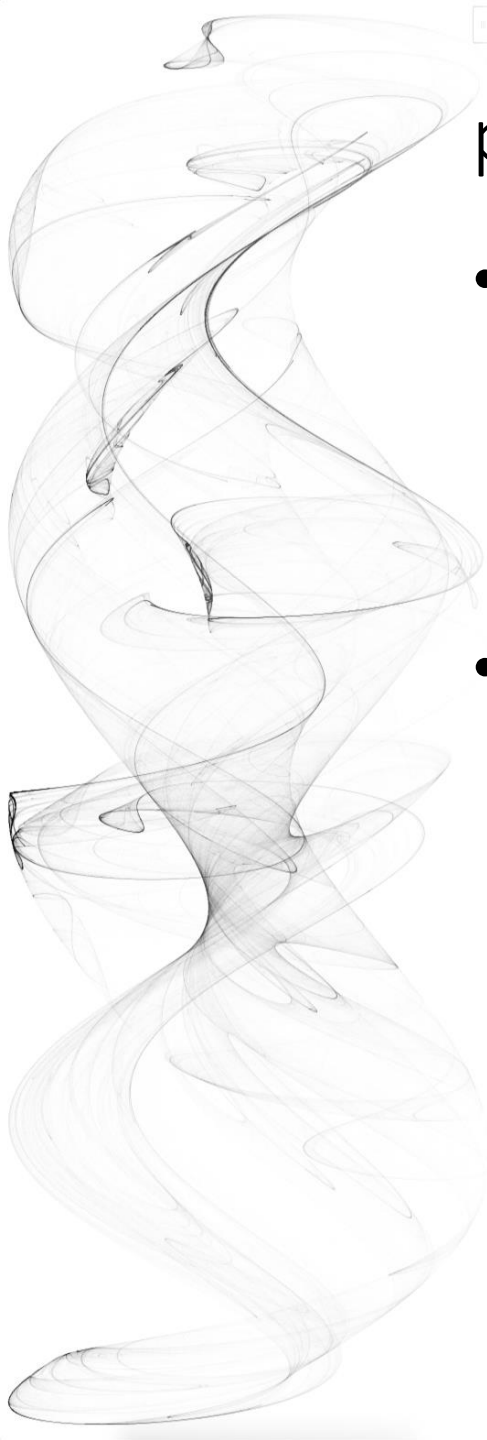
In this case, there is actually a large chance (34%) of seeing **52 or larger under the null hypothesis**. To be more specific, this is a one-sided **right tail test**



p-value

$$z = \frac{\bar{X} - 0.5}{SE(\bar{X})} = \frac{0.02}{\sqrt{\frac{0.5 * (1 - 0.5)}{100}}} = 0.4$$

- Key assumption: in this calculation, we assumed that the observed sample mean should follow a normal distribution (CLT).
- Z-score here, therefore, should follow a standard normal.
- When calculating p-values, we always need to assume an underlying distribution (or approximate distribution) for the quantity that is being tested.



p-value

- p-value shows how likely is a value when the null hypothesis is true
 - $p \leq 0.05$, unlikely
 - $p \leq 0.01$, very unlikely
- A 2014 PNAS paper analyzed success rates from funding agencies in the Netherlands and concluded that their results reveal gender bias favoring male applicants over female applicants in the prioritization of their “quality of researcher” (but not “quality of proposal”) evaluations and success rates, as well as in the language use in instructional and evaluation materials, is the paper statistical solid?

[Know These Problems with P Values | by Andrew Plummer | Towards Data Science](#), also in Moodle

p-value

R 4.2.2 · ~/

```
> library(tidyverse)
> library(dslabs)
> data("research_funding_rates")
> research_funding_rates |>
+ select(discipline, applications_total,
+       success_rates_total) |> head()
```

	discipline	applications_total	success_rates_total
1	Chemical sciences	122	26.2
2	Physical sciences	174	20.1
3	Physics	76	26.3
4	Humanities	396	16.4
5	Technical sciences	251	17.1
6	Interdisciplinary	183	15.8

R 4.2.2 · ~/

```
> names(research_funding_rates)
[1] "discipline"      "applications_total" "applications_men"
[4] "applications_women" "awards_total"      "awards_men"
[7] "awards_women"    "success_rates_total" "success_rates_men"
[10] "success_rates_women"
```

R 4.2.2 · ~/

```
> totals <- research_funding_rates |>
+ select(-discipline) |>
+ summarize_all(sum) |>
+ summarize(yes_men = awards_men,
+          no_men = applications_men - awards_men,
+          yes_women = awards_women,
+          no_women = applications_women - awards_women)
> totals |> summarize(percent_men = yes_men/(yes_men+no_men),
+                   percent_women = yes_women/(yes_women+no_women))
  percent_men percent_women
1    0.17737    0.1489899
```

Use the research_funding_rates dataset

On the left are three columns showing the overall outcomes

We have these values for each gender

We compute the totals that were successful and the totals that were not, we see that a larger percent of men than women received awards

But could this be due just to random variability?

Chi-square test

A chi-square test is a statistical test that is used to compare observed and expected results. The goal of this test is to identify whether a disparity between actual and predicted data is due to chance or to a link between the variables under consideration. The result is a p-value

In our case, we use Chi-square test to answer "is the disparity in successful application rate between men and women because of random variability?"

```
> two_by_two <- data.frame(awarded = c("no", "yes"),  
+                           men = c(totals$no_men, totals$yes_men),  
+                           women = c(totals$no_women, totals$yes_women))
```

```
> two_by_two  
  awarded men women  
1      no 1345  1011  
2      yes  290   177
```

R 4.2.2 · ~/

```
> rate <- totals |>  
+   summarize(percent_total =  
+             (yes_men + yes_women)/  
+             (yes_men + no_men + yes_women + no_women)) |>  
+   pull(percent_total)  
> rate  
[1] 0.1654269  
> data.frame(awarded = c("no", "yes"),  
+             men = (totals$no_men + totals$yes_men) * c(1 - rate, rate),  
+             women = (totals$no_women + totals$yes_women) * c(1 - rate, rate))  
  awarded      men      women  
1      no 1364.5271  991.4729  
2      yes  270.4729  196.5271
```

The first step is to create the two-by-two data table of observations

The general idea of the Chi-square test is to compare the two-by-two table above to what is expected to see if it's totally fair. rate is the rate of assigning funding at random among men and women

Chi-square test

```
R 4.2.2 · ~/   
> chisq_test <- two_by_two |> select(-awarded) |> chisq.test()  
> chisq_test$p.value  
[1] 0.05091372
```

The R function `chisq.test` takes a two-by-two table and returns the results from the test



[Tell your students: Null results are significant too \(rogierkievit.com\)](http://rogierkievit.com)



Larger samples, smaller p-values

- Some studies having large sample sizes tend to report impressively small p-values. Yet, the actual effect size might be modest.
- Same effect size, slightly larger data:

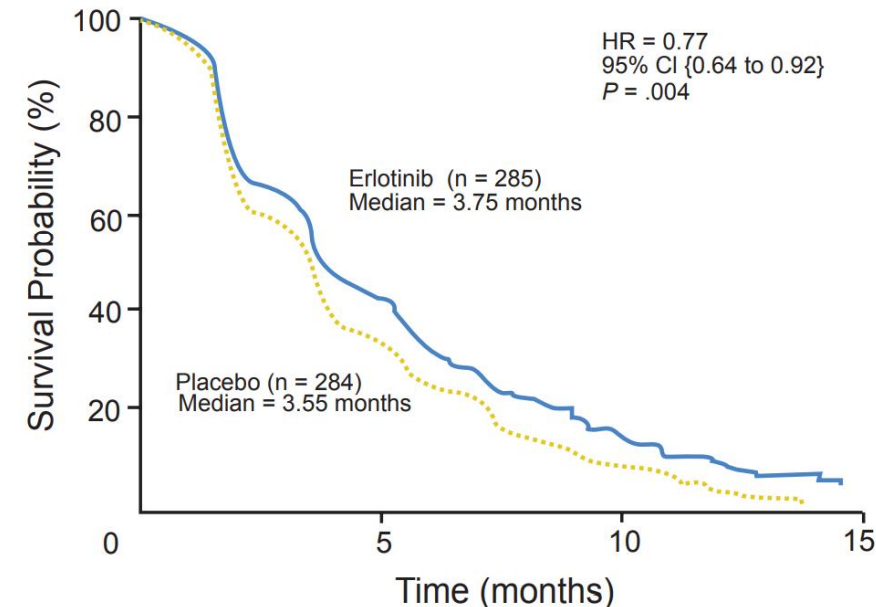
```
> two_by_two_larger <- two_by_two |>
+   mutate(men = as.integer(men * 1.5),
+          women = as.integer(women * 1.5))
> chisq_test <- two_by_two_larger |> select(-awarded) |> chisq.test()
> chisq_test$p.value
[1] 0.01501988
```

p-value versus effect size

- In the extreme case, even tiny effect could be statistically significant with enough amount of data points

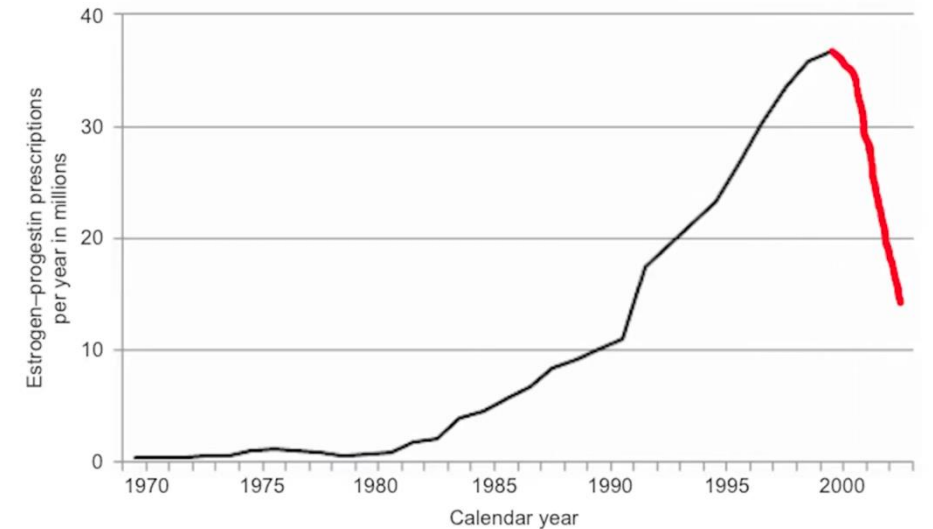
```
> two_by_two_table <- data.frame(A = c(1e6, 2e5), B = c(9.98e5, 2.02e5))
> two_by_two_table
      A      B
1 1e+06 998000
2 2e+05 202000
> chisq.test(two_by_two_table)$p.value
[1] 0.0005493259
> logor <- log((1e6 * 2.02e5)/(2e5 * 9.98e5))
> logor
[1] 0.01195233
```

- GWAS: a variant might be associated with a trait with $p < 10^{-20}$, but only explained 1% of the variance/the trait itself.
- Clinical trial: an anti-cancer drug that significantly extend disease-free survival by 0.2 month



Large samples with a little bit of bias

- In 1980s, large observational studies show that postmenopausal women on hormone replacement therapies had lower CVD risk.
- However, this was later found to be confounded by other factors; women on therapies are wealthier and more health-conscious.
- Randomized controlled trials later in 1990s and 2000s show that such therapies might increase CVD risks.



To learn more: <https://www.youtube.com/watch?v=MVYWqWu2Za4>